

REVIEW

WORKFORCE ANALYTICS: EMPLOYEE ATTRITION ANALYSIS

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Domain: Human Resource Analytics

Introduction

Employee attrition is one of the key challenges faced by organizations as it impacts productivity, operational costs, and workforce stability. Understanding the factors that influence employee turnover is essential for improving employee retention and organizational performance. This project focuses on preparing and cleaning employee-related data to ensure its quality and reliability for future attrition analysis. The study aims to establish a strong data foundation that supports meaningful insights and informed business decisions.

Problem Statement

Organizations often struggle to identify the factors contributing to employee attrition due to the presence of poor-quality data, including missing values, duplicate records, inconsistencies, and outliers. These issues can affect the accuracy of analysis and decision-making. This project aims to assess, clean, and standardize employee data to create a reliable dataset that can be used for future attrition analysis and workforce retention strategies.

Project Objectives

- Understand the employee dataset.
- Assess data quality issues.
- Perform data cleaning and preprocessing.
- Import cleaned data into PostgreSQL.
- Perform SQL-based attrition analysis.
- Generate business insights.
- Prepare data for Power BI dashboard development.

Dataset Overview

- Dataset Name: Employee Attrition Dataset
- Number of Records: 1225
- Number of Attributes: 15
- Target Variable: Attrition
- Granularity: One row represents one employee.

Data Dictionary

Attribute	Description
EmployeeID	Employee Identifier
EmployeeName	Employee Name
Department	Employee Department
JobRole	Employee Role
Gender	Employee Gender
Age	Employee Age
MonthlyIncome	Employee Monthly Income
YearsAtCompany	Employee Experience
OverTime	Overtime Status
JobSatisfaction	Satisfaction Rating
WorkLifeBalance	Work-Life Balance Rating
EnvironmentSatisfaction	Environment Satisfaction Rating
DistanceFromHome	Distance from Home
MaritalStatus	Marital Status
Attrition	Employee Attrition Status

DATA CLEANING AND PREPROCESSING

Dataset Loading

FileHomeTransformAdd ColumnView

Queries [1]

Hr_analysis

= Table.TransformColumnTypes(Source,({"EmployeeID", type text}, {"EmployeeName", type text}, {"Department", type text}, {"JobRole", type

	EmployeeID	EmployeeName	Department	JobRole	Gender	Age	MonthlyIncome	YearsAtCompany	OverTime
1	EMP0001	Jodi Diaz	Marketing	Marketing Executive	Male	38	10524	7	Yes
2	EMP0002	Jesse Robbins	IT	Software Engineer	Female	53	22226	0	Yes
3	EMP0003	Terri Huynh	IT	Software Engineer	female	48	null	8	Yes
4	EMP0004	Adam Green	HR	Recruiter	Male	50	20071	3	No
5	EMP0005	Dawn Jackson	Marketing	Marketing Executive	Male	35	11982	2	Yes
6	EMP0006	Amanda Powers	Sales	Sales Manager	M	34	24460	8	Yes
7	EMP0007	Jessica Donovan	HR	HR Executive	female	41	4332	7	Yes
8	EMP0008	Mark Smith	Operations	Operations Manager	Female	52	15464	20	No
9	EMP0009	Alan Brock	HR	Recruiter	male	46	14361	7	Yes
10	EMP0010	Deborah Pacheco	Sales	Sales Manager	male	25	15108	12	No
11	EMP0011	Lindsey Herman	Marketing	Digital Marketer	female	42	6155	9	No
12	EMP0012	Sarah Tucker	Operations	Operations Executive	female	40	23437	16	Yes
13	EMP0013	David Robbins	Operations	Operations Manager	F	22	6165	11	No
14	EMP0014	Dr. Joshua Johns	Finance	Financial Analyst	male	29	6707	15	Yes
15	EMP0015	Robert Washington	Marketing	Marketing Executive	female	40	15574	20	No
16	EMP0016	Jeffrey Brown	HR	HR Executive	male	56	null	18	Yes
17	EMP0017	Michael Gardner	HR	HR Executive	male	36	null	15	Yes
18	EMP0018	Christina Evans	Finance	Accountant	Female	27	5676	13	No
19	EMP0019	David Butler	Marketing	Marketing Executive	Male	46	13618	3	Yes
20	EMP0020	Ryan Hart	HR	Recruiter	Female	25	17020	17	Yes
21	EMP0021	Steven Rodriguez	Finance	Accountant	Female	46	4421	5	No
22	EMP0022	Michael Baker	Marketing	Digital Marketer	Female	33	12222	6	Yes
23	EMP0023	Kendra Mann	Finance	Financial Analyst	Male	53	5125	5	Yes
24	EMP0024	Eduardo Avery	Operations	Operations Executive	male	59	3802	19	Yes
25	EMP0025	Andrea Beck	Sales	Sales Manager	Female	37	15469	4	No

Description

- Employee attrition dataset was loaded into Excel and Power Query.
- Dataset structure and attributes were reviewed.
- Initial data assessment was performed.

Add Column View

= Table.TransformColumnTypes(Source,({"EmployeeID", type text}, {"EmployeeName", type text}, {"Department", type text}, {"JobRole", type text}, {"Gender", type text}, {"Age", type number}, {"MonthlyIncome", type number}, {"YearsAtCompany", type number}, {"OverTime", type boolean}))

	EmployeeID	EmployeeName	Department	JobRole	Gender	Age	MonthlyIncome	YearsAtCompany	OverTime
1	EMP0001	Jodi Diaz	Marketing	Marketing Executive	Male	38	10524	7	Yes
2	EMP0002	Jesse Robbins	IT	Software Engineer	Female	53	22226	0	Yes
3	EMP0003	Terri Huynh	IT	Software Engineer	female	48	20071	8	Yes
4	EMP0004	Adam Green	HR	Recruiter	Male	50	20071	3	No
5	EMP0005	Dawn Jackson	Marketing	Marketing Executive	Male	35	11982	2	Yes
6	EMP0006	Amanda Powers	Sales	Sales Manager	M	34	24460	8	Yes
7	EMP0007	Jessica Donovan	HR	HR Executive	female	41	4332	7	Yes
8	EMP0008	Mark Smith	Operations	Operations Manager	Female	52	15464	20	No
9	EMP0009	Alan Brock	HR	Recruiter	male	46	14361	7	Yes
10	EMP0010	Deborah Pacheco	Sales	Sales Manager	male	25	15108	12	No
11	EMP0011	Lindsey Herman	Marketing	Digital Marketer	female	42	6155	9	No
12	EMP0012	Sarah Tucker	Operations	Operations Executive	female	40	23437	16	Yes
13	EMP0013	David Robbins	Operations	Operations Manager	F	22	6165	11	No
14	EMP0014	Dr. Joshua Johns	Finance	Financial Analyst	male	29	6707	15	Yes
15	EMP0015	Robert Washington	Marketing	Marketing Executive	female	40	15574	20	No
16	EMP0016	Jeffrey Brown	HR	HR Executive	male	56	null	18	Yes
17	EMP0017	Michael Gardner	HR	HR Executive	male	36	null	15	Yes
18	EMP0018	Christina Evans	Finance	Accountant	Female	27	5676	13	No
19	EMP0019	David Butler	Marketing	Marketing Executive	Male	46	13618	3	Yes
20	EMP0020	Ryan Hart	HR	Recruiter	Female	25	17020	17	Yes
21	EMP0021	Steven Rodriguez	Finance	Accountant	Female	46	4421	5	No
22	EMP0022	Michael Baker	Marketing	Digital Marketer	Female	33	12222	6	Yes
23	EMP0023	Kendra Mann	Finance	Financial Analyst	Male	53	5125	5	Yes
24	EMP0024	Eduardo Avery	Operations	Operations Executive	male	59	3802	19	Yes
25	EMP0025	Andrea Beck	Sales	Sales Manager	Female	37	15469	4	No

- Dataset was profiled to understand attribute distributions.
- Data types and quality issues were examined.
- Potential cleaning requirements were identified.

Cleaned_Employee_Attrition_Synthetic_Dataset - Excel

File Home Insert Draw Page Layout Formulas Data Review View Help Acrobat Table Design

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B I U [Font Icons] [Color Picker]

Font Alignment Number

General [Number Icons]

Conditional Formatting [Table Icons] Cell Styles [Insert Icons]

Insert Delete Format

Σ [Sort & Filter Icons] Find & Select Add-ins Create a PDF

Editing Add-ins Adobe Acrobat

113 [Formula Bar]

	A	B	C	D	E	F	G	H	I	J	K	L
	EmployeeID	EmployeeName	Department	JobRole	Gender	Age	MonthlyIncome	YearsAtCompany	OverTime	JobSatisfaction	WorkLifeBalance	Environment
1	EMP0001	Jodi Diaz	Marketing	Marketing Executive	Male	38	10524	7	Yes		1	5
2	EMP0002	Jesse Robbins	IT	Software Engineer	Female	53	22226	0	Yes			4
3	EMP0003	Terri Huynh	IT	Software Engineer	female	48		8	Yes			3
4	EMP0004	Adam Green	HR	Recruiter	Male	50	20071	3	No		1	5
5	EMP0005	Dawn Jackson	Marketing	Marketing Executive	Male	35	11982	2	Yes		1	4
6	EMP0006	Amanda Powers	Sales	Sales Manager	M	34	24460	8	Yes			2
7	EMP0007	Jessica Donovan	HR	HR Executive	female	41	4332	7	Yes			4
8	EMP0008	Mark Smith	Operations	Operations Manager	Female	52	15464	20	No		2	3
9	EMP0009	Alan Brock	HR	Recruiter	male	46	14361	7	Yes		5	4
10	EMP0010	Deborah Pacheco	Sales	Sales Manager	male	25	15108	12	No		5	3
11	EMP0011	Lindsey Herman	Marketing	Digital Marketer	female	42	6155	9	No		2	4
12	EMP0012	Sarah Tucker	Operations	Operations Executive	female	40	23437	16	Yes		2	3
13	EMP0013	David Robbins	Operations	Operations Manager	F	22	6165	11	No		2	1
14	EMP0014	Dr. Joshua Johns	Finance	Financial Analyst	male	29	6707	15	Yes		3	5
15	EMP0015	Robert Washington	Marketing	Marketing Executive	female	40	15574	20	No		4	5
16	EMP0016	Jeffrey Brown	HR	HR Executive	male	56		18	Yes		1	1
17	EMP0017	Michael Gardner	HR	HR Executive	male	36		15	Yes		5	2
18	EMP0018	Christina Evans	Finance	Accountant	Female	27	5676	13	No		4	4
19	EMP0019	David Butler	Marketing	Marketing Executive	Male	46	13618	3	Yes		2	2
20	EMP0020	Ryan Hart	HR	Recruiter	Female	25	17020	17	Yes		1	5

Raw employee attrition analysis Cleaned_Employee_Attrition_Synt

- Missing values were identified in multiple attributes.
- Missing data can impact analytical accuracy.
- Missing values were documented for treatment.

Removed duplicates:

Queries [1]

Hr_analysis

×

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f_x

= Table.Distinct("#Changed Type")

1

EMP0001

Jodi Diaz

Marketing

Marketing Executive

Male

38

2

EMP0002

Jesse Robbins

IT

Software Engineer

Female

53

3

EMP0003

Terri Huynh

IT

Software Engineer

female

48

4

EMP0004

Adam Green

HR

Recruiter

Male

50

5

EMP0005

Dawn Jackson

Marketing

Marketing Executive

Male

35

6

EMP0006

Amanda Powers

Sales

Sales Manager

M

34

7

EMP0007

Jessica Donovan

HR

HR Executive

female

41

8

EMP0008

Mark Smith

Operations

Operations Manager

Female

52

9

EMP0009

Alan Brock

HR

Recruiter

male

46

10

EMP0010

Deborah Pacheco

Sales

Sales Manager

male

25

11

EMP0011

Lindsey Herman

Marketing

Digital Marketer

female

42

12

EMP0012

Sarah Tucker

Operations

Operations Executive

female

40

13

EMP0013

David Robbins

Operations

Operations Manager

F

22

14

EMP0014

Dr. Joshua Johns

Finance

Financial Analyst

male

29

15

EMP0015

Robert Washington

Marketing

Marketing Executive

female

40

16

EMP0016

Jeffrey Brown

HR

HR Executive

male

56

17

EMP0017

Michael Gardner

HR

HR Executive

male

36

18

EMP0018

Christina Evans

Finance

Accountant

Female

27

19

EMP0019

David Butler

Marketing

Marketing Executive

Male

46

20

EMP0020

Ryan Hart

HR

Recruiter

Female

25

21

EMP0021

Steven Rodriguez

Finance

Accountant

Female

46

22

EMP0022

Michael Baker

Marketing

Digital Marketer

Female

33

23

EMP0023

Kendra Mann

Finance

Financial Analyst

Male

53

24

EMP0024

Eduardo Avery

Operations

Operations Executive

male

59

25

EMP0025

Andrea Beck

Sales

Sales Manager

Female

37

26

Query Settings

PROPERTIES

Name

Hr_analysis

All Properties

APPLIED STEPS

Source

Changed Type

Removed Duplicates

15 COLUMNS, 999+ ROWS

PREVIEW DOWNLOADED AT 19:01

Description

- The dataset was examined for duplicate employee records.
- Duplicate entries can lead to inaccurate employee counts and misleading analytical results.
- All identified duplicate records were removed to maintain data integrity and consistency.

Data Standardization Capitalization:

Queries [1] fx = Table.TransformColumns(#"Removed Duplicates",{{"EmployeeName", Text.Proper, type text},

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170	171	172	173	174	175	176	177	178	179	180	181	182	183	184	185	186	187	188	189	190	191	192	193	194	195	196	197	198	199	200	201	202	203	204	205	206	207	208	209	210	211	212	213	214	215	216	217	218	219	220	221	222	223	224	225	226	227	228	229	230	231	232	233	234	235	236	237	238	239	240	241	242	243	244	245	246	247	248	249	250	251	252	253	254	255	256	257	258	259	260	261	262	263	264	265	266	267	268	269	270	271	272	273	274	275	276	277	278	279	280	281	282	283	284	285	286	287	288	289	290	291	292	293	294	295	296	297	298	299	300	301	302	303	304	305	306	307	308	309	310	311	312	313	314	315	316	317	318	319	320	321	322	323	324	325	326	327	328	329	330	331	332	333	334	335	336	337	338	339	340	341	342	343	344	345	346	347	348	349	350	351	352	353	354	355	356	357	358	359	360	361	362	363	364	365	366	367	368	369	370	371	372	373	374	375	376	377	378	379	380	381	382	383	384	385	386	387	388	389	390	391	392	393	394	395	396	397	398	399	400	401	402	403	404	405	406	407	408	409	410	411	412	413	414	415	416	417	418	419	420	421	422	423	424	425	426	427	428	429	430	431	432	433	434	435	436	437	438	439	440	441	442	443	444	445	446	447	448	449	450	451	452	453	454	455	456	457	458	459	460	461	462	463	464	465	466	467	468	469	470	471	472	473	474	475	476	477	478	479	480	481	482	483	484	485	486	487	488	489	490	491	492	493	494	495	496	497	498	499	500	501	502	503	504	505	506	507	508	509	510	511	512	513	514	515	516	
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Description

- Text values were standardized using capitalization rules.
- Inconsistent text formats were corrected to improve data consistency.
- Standardized values help ensure accurate grouping and reporting during analysis.

Gender column

Queries [1]

Hr_analysis

= Table.ReplaceValue("#Replaced Value","Female","F",Replacer.ReplaceText,{"Gender"})

	EmployeeID	EmployeeName	Department	JobRole	Gender	Age	MonthlyIncome
1	EMP0001	Jodi Diaz	Marketing	Marketing Executive	M	38	
2	EMP0002	Jesse Robbins	It	Software Engineer	F	53	
3	EMP0003	Terri Huynh	It	Software Engineer	F	48	
4	EMP0004	Adam Green	Hr	Recruiter	M	50	
5	EMP0005	Dawn Jackson	Marketing	Marketing Executive	M	35	
6	EMP0006	Amanda Powers	Sales	Sales Manager	M	34	
7	EMP0007	Jessica Donovan	Hr	Hr Executive	F	41	
8	EMP0008	Mark Smith	Operations	Operations Manager	F	52	
9	EMP0009	Alan Brock	Hr	Recruiter	M	46	
10	EMP0010	Deborah Pacheco	Sales	Sales Manager	M	25	
11	EMP0011	Lindsey Herman	Marketing	Digital Marketer	F	42	
12	EMP0012	Sarah Tucker	Operations	Operations Executive	F	40	
13	EMP0013	David Robbins	Operations	Operations Manager	F	22	
14	EMP0014	Dr. Joshua Johns	Finance	Financial Analyst	M	29	
15	EMP0015	Robert Washington	Marketing	Marketing Executive	F	40	
16	EMP0016	Jeffrey Brown	Hr	Hr Executive	M	56	
17	EMP0017	Michael Gardner	Hr	Hr Executive	M	36	
18	EMP0018	Christina Evans	Finance	Accountant	F	27	
19	EMP0019	David Butler	Marketing	Marketing Executive	M	46	
20	EMP0020	Ryan Hart	Hr	Recruiter	F	25	
21	EMP0021	Steven Rodriguez	Finance	Accountant	F	46	
22	EMP0022	Michael Baker	Marketing	Digital Marketer	F	33	
23	EMP0023	Kendra Mann	Finance	Financial Analyst	M	53	
24	EMP0024	Eduardo Avery	Operations	Operations Executive	M	59	
25	EMP0025	Andrea Beck	Sales	Sales Manager	F	37	

Query Settings

PROPERTIES

Name
Hr_analysis

All Properties

APPLIED STEPS

- Source
- Changed Type
- Removed Duplicates
- Capitalized Each Word
- Replaced Value
- X Replaced Value1

Description

- Inconsistent gender representations were identified in the dataset.
- Values were standardized into a consistent format to avoid duplicate categories.
- This transformation improves the accuracy of gender-based analysis and reporting.

Age Column

Queries | **Hr_analysis**

Table.SelectRows("#Replaced Value1", each [Age] < 65 and [Age] > 18)

	EmployeeID	EmployeeName	Department	JobRole	Gender	Age	MonthlyIncome
1	EMP0001	Jodi Diaz	Marketing	Marketing Executive	M	38	
2	EMP0002	Jesse Robbins	It	Software Engineer	F	53	
3	EMP0003	Terri Huynh	It	Software Engineer	F	48	
4	EMP0004	Adam Green	Hr	Recruiter	M	50	
5	EMP0005	Dawn Jackson	Marketing	Marketing Executive	M	35	
6	EMP0006	Amanda Powers	Sales	Sales Manager	M	34	
7	EMP0007	Jessica Donovan	Hr	Hr Executive	F	41	
8	EMP0008	Mark Smith	Operations	Operations Manager	F	52	
9	EMP0009	Alan Brock	Hr	Recruiter	M	46	
10	EMP0010	Deborah Pacheco	Sales	Sales Manager	M	25	
11	EMP0011	Lindsey Herman	Marketing	Digital Marketer	F	42	
12	EMP0012	Sarah Tucker	Operations	Operations Executive	F	40	
13	EMP0013	David Robbins	Operations	Operations Manager	F	22	
14	EMP0014	Dr. Joshua Johns	Finance	Financial Analyst	M	29	
15	EMP0015	Robert Washington	Marketing	Marketing Executive	F	40	
16	EMP0016	Jeffrey Brown	Hr	Hr Executive	M	56	
17	EMP0017	Michael Gardner	Hr	Hr Executive	M	36	
18	EMP0018	Christina Evans	Finance	Accountant	F	27	
19	EMP0019	David Butler	Marketing	Marketing Executive	M	46	
20	EMP0020	Ryan Hart	Hr	Recruiter	F	25	
21	EMP0021	Steven Rodriguez	Finance	Accountant	F	46	
22	EMP0022	Michael Baker	Marketing	Digital Marketer	F	33	
23	EMP0023	Kendra Mann	Finance	Financial Analyst	M	53	
24	EMP0024	Eduardo Avery	Operations	Operations Executive	M	59	
25	EMP0025	Andrea Beck	Sales	Sales Manager	F	37	

Query Settings

- PROPERTIES**
 - Name: Hr_analysis
 - All Properties
- APPLIED STEPS**
 - Source
 - Changed Type
 - Removed Duplicates
 - Capitalized Each Word
 - Replaced Value
 - Replaced Value1
 - Filtered Rows**

Description

- Invalid age values were identified during data quality assessment.
- Employee records with age values outside the acceptable business range were treated as outliers.
- Outlier records were removed using filtering techniques to improve data reliability.

Monthly income:

Queries [1]

Hr_analysis

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fx

Table.SelectRows("#Filtered Rows", each ([MonthlyIncome] = null))

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JobRole

Gender

Age

MonthlyIncome

YearsAtCompany

OverTime

1	Software Engineer	F	48	null	8	Yes
2	Hr Executive	M	56	null	18	Yes
3	Hr Executive	M	36	null	15	Yes
4	Sales Manager	M	32	null	13	Yes
5	Sales Executive	F	22	null	0	No
6	Operations Manager	F	45	null	11	Yes
7	Financial Analyst	F	45	null	8	Yes
8	Recruiter	F	29	null	16	Yes
9	Sales Manager	M	21	null	18	Yes
10	Digital Marketer	F	27	null	16	Yes
11	Financial Analyst	M	37	null	20	No
12	Data Analyst	F	33	null	17	No
13	Operations Executive	M	52	null	1	Yes
14	Qa Engineer	F	28	null	16	Yes
15	Accountant	M	47	null	17	No
16	Accountant	M	53	null	12	No
17	Qa Engineer	F	57	null	10	Yes
18	Operations Executive	M	44	null	1	No
19	Sales Executive	M	49	null	19	Yes
20	Marketing Executive	F	45	null	10	Yes
21	Digital Marketer	M	55	null	6	Yes
22	Operations Manager	F	30	null	20	No
23	Hr Executive	F	58	null	10	No
24	Accountant	M	23	null	11	No
25	Financial Analyst	F	60	null	14	Yes

Query Settings

PROPERTIES

Name

Hr_analysis

All Properties

APPLIED STEPS

Source

Changed Type

Removed Duplicates

Capitalized Each Word

Replaced Value

Replaced Value1

Filtered Rows

Filtered Rows1

Description

- Missing values were identified in the Monthly Income attribute.
- Null values can negatively affect statistical analysis and reporting.
- Appropriate treatment was required to maintain dataset completeness.

Removed null values by median imputation:

Calculated median:

Queries [1]

123 Hr_analysis

13935

✕

✓

$\frac{fx}{x}$

= List.Median(#"Filtered Rows"[MonthlyIncome])

▼

Query Settings

✕

▲ PROPERTIES

Name

Hr_analysis

All Properties

▲ APPLIED STEPS

Source

Changed Type

Removed Duplicates

Capitalized Each Word

Replaced Value

Replaced Value1

Filtered Rows

✕ Calculated Median

Queries [1] ✕ ✓ fx = Table.ReplaceValue("#Filtered Rows",null,13935,Replacer.ReplaceValue,{"MonthlyIncome"})

Hr_analysis

	JobRole	Gender	Age	MonthlyIncome	YearsAtCompany	OverTime	JobSatisf
1	Marketing Executive	M	38	10524	7	Yes	
2	Software Engineer	F	53	22226	0	Yes	
3	Software Engineer	F	48	13935	8	Yes	
4	Recruiter	M	50	20071	3	No	
5	Marketing Executive	M	35	11982	2	Yes	
6	Sales Manager	M	34	24460	8	Yes	
7	Hr Executive	F	41	4332	7	Yes	
8	Operations Manager	F	52	15464	20	No	
9	Recruiter	M	46	14361	7	Yes	
10	Sales Manager	M	25	15108	12	No	
11	Digital Marketer	F	42	6155	9	No	
12	Operations Executive	F	40	23437	16	Yes	
13	Operations Manager	F	22	6165	11	No	
14	Financial Analyst	M	29	6707	15	Yes	
15	Marketing Executive	F	40	15574	20	No	
16	Hr Executive	M	56	13935	18	Yes	
17	Hr Executive	M	36	13935	15	Yes	
18	Accountant	F	27	5676	13	No	
19	Marketing Executive	M	46	13618	3	Yes	
20	Recruiter	F	25	17020	17	Yes	
21	Accountant	F	46	4421	5	No	
22	Digital Marketer	F	33	12222	6	Yes	
23	Financial Analyst	M	53	5125	5	Yes	
24	Operations Executive	M	59	3802	19	Yes	
25	Sales Manager	F	37	15469	4	No	

Query Settings ✕

PROPERTIES

Name
Hr_analysis

All Properties

APPLIED STEPS

Source
Changed Type
Removed Duplicates
Capitalized Each Word
Replaced Value
Replaced Value1
Filtered Rows
Replaced Value2

Description

- The median value of Monthly Income was calculated.
- Missing income values were replaced using median imputation.
- Median was selected because it is less affected by extreme income values and provides a more representative measure of central tendency.

Job satisfaction:

Replaced by mean and rounded of to 3

Queries [1] ✕ ✓ fx = List.Average("#Replaced Value2"[JobSatisfaction])

Hr_analysis

2.963555555555557

Query Settings ✕

PROPERTIES

Name
Hr_analysis

All Properties

APPLIED STEPS

Source
Changed Type
Removed Duplicates
Capitalized Each Word
Replaced Value
Replaced Value1
Filtered Rows
Replaced Value2
Calculated Average

Description

- Missing values were identified in the JobSatisfaction attribute.
- The average value was calculated and rounded to the nearest whole number (3).
- Missing values were replaced with the rounded average value to maintain data consistency.

Work life balance:

Replaced by average and rounded off to 3

Queries [1]
Hr_analysis

3.048

Formula Bar: `= List.Average(#"Replaced Value3"[WorkLifeBalance])`

Query Settings

PROPERTIES

Name: Hr_analysis

All Properties

APPLIED STEPS

- Source
- Changed Type
- Removed Duplicates
- Capitalized Each Word
- Replaced Value
- Replaced Value1
- Filtered Rows
- Replaced Value2
- Replaced Value3
- Calculated Average**

READY

Queries [1]
Hr_analysis

Formula Bar: `= Table.ReplaceValue(#"Replaced Value3",null,3,Replacer.ReplaceValue,{"WorkLifeBalance"})`

	YearsAtCompany	OverTime	JobSatisfaction	WorkLifeBalance	EnvironmentSatisfaction
1	10524	7 Yes	1	5	1
2	22226	0 Yes	3	4	2
3	13935	8 Yes	3	3	1
4	20071	3 No	1	5	3
5	11982	2 Yes	1	4	3
6	24460	8 Yes	3	2	5
7	4332	7 Yes	3	4	3
8	15464	20 No	2	3	2
9	14361	7 Yes	5	4	1
10	15108	12 No	5	3	5
11	6155	9 No	2	4	1
12	23437	16 Yes	2	3	2
13	6165	11 No	2	1	2
14	6707	15 Yes	3	5	5
15	15574	20 No	4	5	4
16	13935	18 Yes	1	1	1
17	13935	15 Yes	5	2	5
18	5676	13 No	4	4	4
19	13618	3 Yes	2	2	5
20	17020	17 Yes	1	5	1
21	4421	5 No	1	4	3
22	12222	6 Yes	5	5	1
23	5125	5 Yes	3	1	2
24	3802	19 Yes	4	5	5
25	15469	4 No	4	3	1

Query Settings

PROPERTIES

Name: Hr_analysis

All Properties

APPLIED STEPS

- Source
- Changed Type
- Removed Duplicates
- Capitalized Each Word
- Replaced Value
- Replaced Value1
- Filtered Rows
- Replaced Value2
- Replaced Value3
- Replaced Value4**

Description

- Missing values were identified in the WorkLifeBalance attribute.
- The average value was calculated and rounded to the nearest whole number (3).
- Missing values were replaced using the rounded average value to preserve the overall distribution of the data.

Final Data Validation

	A	B	C	D	E	F	G	H	I	J	K	L
	EmployeeID	EmployeeName	Department	JobRole	Gender	Age	MonthlyIncome	YearsAtCompany	OverTime	JobSatisfaction	WorkLifeBalance	EnvironmentSatisfaction
2	EMP0001	Jodi Diaz	Marketing	Marketing	M	38	10524	7	Yes	1	5	1
3	EMP0002	Jesse Robbins	It	Software E	F	53	22226	0	Yes	3	4	2
4	EMP0003	Terri Huynh	It	Software E	F	48	13935	8	Yes	3	3	1
5	EMP0004	Adam Green	Hr	Recruiter	M	50	20071	3	No	1	5	3
6	EMP0005	Dawn Jackson	Marketing	Marketing	M	35	11982	2	Yes	1	4	3
7	EMP0006	Amanda Powers	Sales	Sales Man	M	34	24460	8	Yes	3	2	5
8	EMP0007	Jessica Donovan	Hr	Hr Executiv	F	41	4332	7	Yes	3	4	3
9	EMP0008	Mark Smith	Operations	Operations	F	52	15464	20	No	2	3	2
10	EMP0009	Alan Brock	Hr	Recruiter	M	46	14361	7	Yes	5	4	1
11	EMP0010	Deborah Pacheco	Sales	Sales Man	M	25	15108	12	No	5	3	5
12	EMP0011	Lindsey Herman	Marketing	Digital Mar	F	42	6155	9	No	2	4	1
13	EMP0012	Sarah Tucker	Operations	Operations	F	40	23437	16	Yes	2	3	2
14	EMP0013	David Robbins	Operations	Operations	F	22	6165	11	No	2	1	2
15	EMP0014	Dr. Joshua Johns	Finance	Financial A	M	29	6707	15	Yes	3	5	5
16	EMP0015	Robert Washington	Marketing	Marketing	F	40	15574	20	No	4	5	4
17	EMP0016	Jeffrey Brown	Hr	Hr Executiv	M	56	13935	18	Yes	1	1	1
18	EMP0017	Michael Gardner	Hr	Hr Executiv	M	36	13935	15	Yes	5	2	5
19	EMP0018	Christina Evans	Finance	Accountan	F	27	5676	13	No	4	4	4
20	EMP0019	David Butler	Marketing	Marketing	M	46	13618	3	Yes	2	2	5
21	EMP0020	Ryan Hart	Hr	Recruiter	F	25	17020	17	Yes	1	5	1

Description

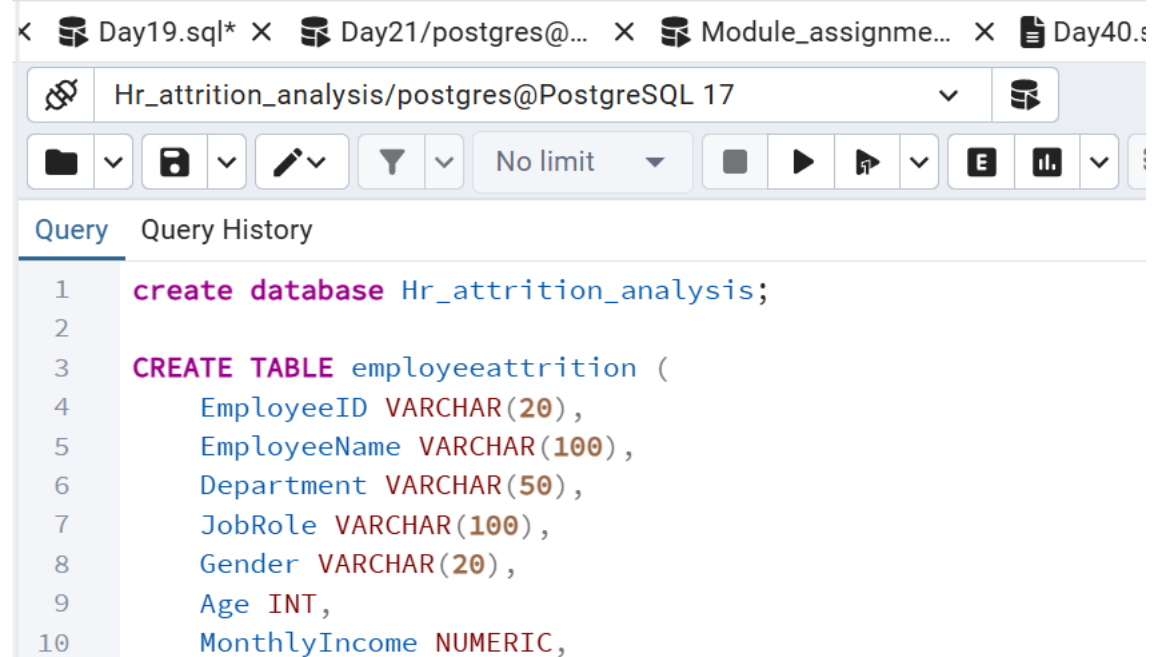
- The cleaned dataset was reviewed to verify all transformations.
- Missing values, duplicate records, inconsistencies, and outliers were successfully addressed.
- The dataset was validated and prepared for SQL-based analysis.

Export of Cleaned Dataset

- Cleaned dataset was exported as CSV format.
- CSV file was prepared for PostgreSQL import.
- Data was validated before import.

SQL ANALYSIS

PostgreSQL Database Creation



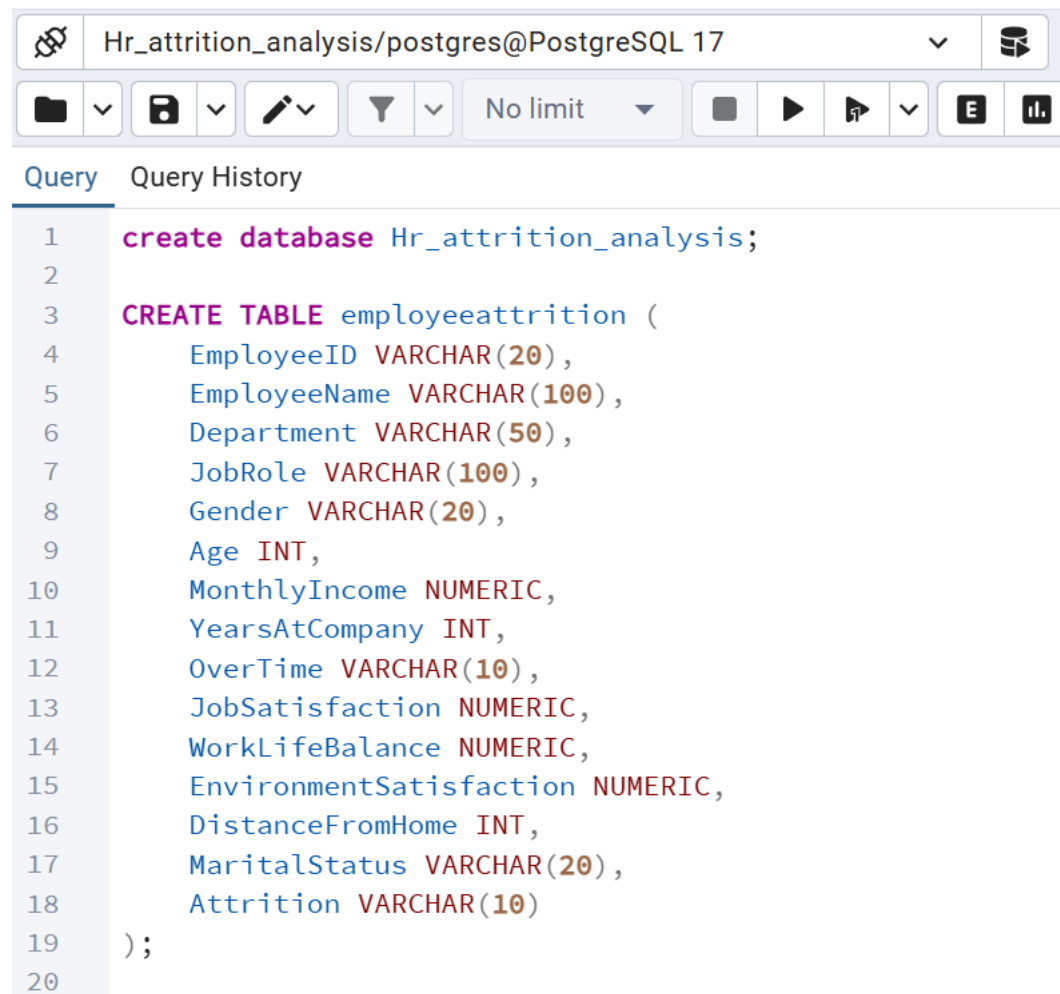
The screenshot shows a PostgreSQL query editor interface. At the top, there are several tabs: 'Day19.sql*', 'Day21/postgres@...', 'Module_assignme...', and 'Day40.s'. The active tab is 'Hr_attrition_analysis/postgres@PostgreSQL 17'. Below the tabs is a toolbar with icons for file operations, query execution, and other database functions. The main area displays a SQL query with line numbers 1 through 10. The query consists of two statements: 'create database Hr_attrition_analysis;' and 'CREATE TABLE employeattrition (' followed by a list of columns and their data types: EmployeeID VARCHAR(20), EmployeeName VARCHAR(100), Department VARCHAR(50), JobRole VARCHAR(100), Gender VARCHAR(20), Age INT, and MonthlyIncome NUMERIC.

```
1 create database Hr_attrition_analysis;
2
3 CREATE TABLE employeattrition (
4     EmployeeID VARCHAR(20),
5     EmployeeName VARCHAR(100),
6     Department VARCHAR(50),
7     JobRole VARCHAR(100),
8     Gender VARCHAR(20),
9     Age INT,
10    MonthlyIncome NUMERIC,
```

Description

- HR_Attrition_Analysis database was created.
- Database environment was configured.
- Prepared for storing employee data.

16. Table Creation



The screenshot shows a PostgreSQL query editor window. The title bar indicates the connection is 'Hr_attrition_analysis/postgres@PostgreSQL 17'. The interface includes a toolbar with icons for file operations, query execution, and settings. Below the toolbar, there are tabs for 'Query' and 'Query History'. The main area displays a SQL script for creating a database and a table.

```
1  create database Hr_attrition_analysis;
2
3  CREATE TABLE employeeattrition (
4      EmployeeID VARCHAR(20),
5      EmployeeName VARCHAR(100),
6      Department VARCHAR(50),
7      JobRole VARCHAR(100),
8      Gender VARCHAR(20),
9      Age INT,
10     MonthlyIncome NUMERIC,
11     YearsAtCompany INT,
12     OverTime VARCHAR(10),
13     JobSatisfaction NUMERIC,
14     WorkLifeBalance NUMERIC,
15     EnvironmentSatisfaction NUMERIC,
16     DistanceFromHome INT,
17     MaritalStatus VARCHAR(20),
18     Attrition VARCHAR(10)
19 );
20
```

Description

- Employee Attrition table was created.
- Appropriate data types were assigned.
- Table structure was validated.

Data validation

Total records

plorer | Hr_attrition_analysis/postgres@PostgreSQL 17

Query

```
12 WorkLifeBalance NUMERIC,
13 EnvironmentSatisfaction NUMERIC,
14 DistanceFromHome INT,
15 MaritalStatus VARCHAR(20),
16 Attrition VARCHAR(10)
17 );
18
19 SELECT COUNT(*)
20 FROM employeeattrition;
21
22
```

Data Output

	count
1	1197

Showing rows: 1 to 1 | Page N

Total rows: 1 | Query complete 00:00:00.128

First 10 records:

plorer | Hr_attrition_analysis/postgres@PostgreSQL 17

Query

```
21
22 SELECT *
23 FROM employeeattrition
24 LIMIT 10;
```

Data Output

	employeeid	employeenam	department	jobrole	gender	age	monthlyincome	years
	character varying (20)	character varying (100)	character varying (50)	character varying (100)	character varying (20)	integer	numeric	integer
1	EMP0001	Jodi Diaz	Marketing	Marketing Executive	M	38	10524	
2	EMP0002	Jesse Robbins	It	Software Engineer	F	53	22226	
3	EMP0003	Terri Huynh	It	Software Engineer	F	48	13935	
4	EMP0004	Adam Green	Hr	Recruiter	M	50	20071	
5	EMP0005	Dawn Jackson	Marketing	Marketing Executive	M	35	11982	
6	EMP0006	Amanda Powers	Sales	Sales Manager	M	34	24460	
7	EMP0007	Jessica Donovan	Hr	Hr Executive	F	41	4332	
8	EMP0008	Mark Smith	Operations	Operations Manager	F	52	15464	
9	EMP0009	Alan Brock	Hr	Recruiter	M	46	14361	
10	EMP0010	Deborah Pacheco	Sales	Sales Manager	M	25	15108	

Showing rows: 1 to 10 | Page No: 1 | of 1

Total rows: 10 | Query complete 00:00:00.267 | CRLF | Ln 21, Col 1

Total employees:

The screenshot shows a PostgreSQL query editor interface. At the top, the connection is 'Hr_attrition_analysis/postgres@PostgreSQL 17'. A message states 'The session is idle and there is no current transaction.' Below the toolbar, the 'Query' tab is active, displaying a SQL query across lines 21 to 32. The query consists of two parts: a first query limited to 10 rows, and a second query that counts the total number of employees. The second query is highlighted in blue. Below the query editor, the 'Data Output' tab shows the results of the query. It displays a single row with the column 'total_employees' of type 'bigint' and a value of 1197.

```
21
22 SELECT *
23 FROM employeeattrition
24 LIMIT 10;
25
26 SELECT *
27 FROM employeeattrition
28 WHERE jobsatisfaction IS NULL
29 OR worklifebalance IS NULL;
30
31 SELECT COUNT(*) AS total_employees
32 FROM employeeattrition;
```

	total_employees bigint
1	1197

Description

- The total number of employees present in the organization was calculated.
- This analysis provides an overview of the workforce size available in the dataset.
- The result serves as the baseline for all further attrition and workforce analyses.

Employees by gender:

The screenshot shows a PostgreSQL query editor interface. At the top, the connection is 'Hr_attrition_analysis/postgres@PostgreSQL 17'. A message states 'The session is idle and there is no current transaction.' Below this, the 'Query' tab is active, displaying a SQL query. The query is as follows:

```
26 SELECT *
27 FROM employeeattrition
28 WHERE jobsatisfaction IS NULL
29 OR worklifebalance IS NULL;
30
31 SELECT COUNT(*) AS total_employees
32 FROM employeeattrition;
33
34 SELECT gender,
35 COUNT(*) AS employee_count
36 FROM employeeattrition
37 GROUP BY gender;
```

Below the query editor, the 'Data Output' tab is active, showing the results of the query. The results are displayed in a table with two columns: 'gender' (character varying (20)) and 'employee_count' (bigint). The table contains two rows: one for 'M' with a count of 564, and one for 'F' with a count of 633.

	gender character varying (20)	employee_count bigint
1	M	564
2	F	633

Description

- The workforce was analyzed based on gender distribution within the organization.
- This analysis helps understand the representation of different gender groups across the employee population.
- The findings provide insights into workforce diversity and support demographic analysis for organizational planning.

Job Satisfaction Analysis

The screenshot shows a database management interface with a query editor and a results pane. The query editor displays a SQL query that selects job satisfaction levels and counts employees who have left, grouped by job satisfaction. The results pane shows a table with 5 rows of data.

Query:

```
84  
85 SELECT jobsatisfaction,  
86 COUNT(*) AS employees_left  
87 FROM employeeattrition  
88 WHERE attrition='Yes'  
89 GROUP BY jobsatisfaction  
90 ORDER BY jobsatisfaction;  
91  
92  
93
```

Data Output:

	jobsatisfaction numeric	employees_left bigint
1	1	60
2	2	52
3	3	65
4	4	51
5	5	50

Total rows: 5 Query complete 00:00:00.651

Description

- Employee attrition was analyzed based on job satisfaction levels.
- The analysis helps identify the relationship between employee satisfaction and turnover.
- The results support initiatives aimed at improving employee engagement and retention.

Work life balance by attrition

MySQL20/postgre... X Day19.sql* X Day21/postgres@...

Hr_attrition_analysis/postgres@PostgreSQL 17

Query Query History

```
91
92 SELECT worklifebalance,
93 COUNT(*) AS employees_left
94 FROM employeeattrition
95 WHERE attrition='Yes'
96 GROUP BY worklifebalance
97 ORDER BY worklifebalance;
98
99
100
```

Data Output Messages Notifications

	worklifebalance numeric	employees_left bigint
1	1	49
2	2	54
3	3	70
4	4	50
5	5	55

Description

- Employee turnover was examined across different work-life balance ratings.
- The objective was to assess whether work-life balance influences employee retention.
- The findings support organizational efforts to improve employee well-being.

Environment Satisfaction by Attrition

 Hr_attrition_analysis/postgres@PostgreSQL 17

    No limit   

Query

Query History

96

GROUP BY worklifebalance

97

ORDER BY worklifebalance;

98

99

SELECT environmentsatisfaction,

100

COUNT(*) AS employees_left

101

FROM employeeattrition

102

WHERE attrition='Yes'

103

GROUP BY environmentsatisfaction

104

ORDER BY environmentsatisfaction;

105

Data Output

Messages

Notifications

       SQL

	environmentsatisfaction numeric	employees_left bigint
1	1	56
2	2	58
3	3	51
4	4	57
5	5	56

Description

- Employee attrition was analyzed based on workplace environment satisfaction levels.
- The analysis evaluated the impact of workplace conditions on retention.
- The results help identify opportunities to improve employee experience.

Monthly Income Analysis

 Hr_attrition_analysis/postgres@PostgreSQL 17



No limit



Query

Query History

100

COUNT(*) AS employees_left

101

FROM employeeattrition

102

WHERE attrition='Yes'

103

GROUP BY environmentsatisfaction

104

ORDER BY environmentsatisfaction;

105

106

SELECT

107

ROUND(AVG(monthlyincome),2) AS avg_income_left

108

FROM employeeattrition

109

WHERE attrition='Yes';

Data Output

Messages

Notifications



SQL

	avg_income_left 
1	14222.22

Total rows: 1

Query complete 00:00:00.173

Description

- The average monthly income of employees who left the organization was analyzed.
- This analysis helps assess whether compensation plays a significant role in employee attrition.
- The results support decision-making related to salary and employee retention strategies.

Attrition by Salary Band

Hr_attrition_analysis/postgres@PostgreSQL 17

No limit

Query Query History

111 SELECT

112 CASE

113 WHEN monthlyincome < 5000 THEN 'Low Income'

114 WHEN monthlyincome BETWEEN 5000 AND 10000 THEN 'Medium Income'

115 ELSE 'High Income'

116 END AS salary_band,

117 COUNT(*) AS employees_left

118 FROM employeeattrition

119 WHERE attrition='Yes'

120 GROUP BY salary_band;

121

Data Output Messages Notifications

SQL

Showing r

	salary_band text	employees_left bigint
1	Medium Income	59
2	High Income	198
3	Low Income	21

Total rows: 3 Query complete 00:00:00.358

Description

- Employees were categorized into salary bands to study attrition patterns.
- The objective was to identify income groups experiencing higher employee turnover.
- The findings provide insights into compensation-related retention challenges.

Years at Company vs Attrition

MySQL20/postgre... X Day19.sql* X Day21/postgres@... X Module_assignme... X

Hr_attrition_analysis/postgres@PostgreSQL 17

Query Query History

```
119 WHERE attrition='Yes'
120 GROUP BY salary_band;
121
122 SELECT yearsatcompany,
123 COUNT(*) AS employees_left
124 FROM employeeattrition
125 WHERE attrition='Yes'
126 GROUP BY yearsatcompany
127 ORDER BY yearsatcompany;
128
```

Data Output Messages Notifications

Showing rows: 1 to 21

	yearsatcompany integer	employees_left bigint
1	0	8
2	1	14
3	2	20
4	3	11
5	4	12
6	5	7

Total rows: 21 Query complete 00:00:00.189

Description

- Employee attrition was analyzed based on years of service within the organization.
- The analysis helps identify at which stage employees are more likely to leave.
- Understanding tenure-based attrition supports workforce planning and retention programs.

Distance from Home vs Attrition

Hr_attrition_analysis/postgres@PostgreSQL 17

No limit

Query Query History

129

130

131

132

133

134

135

136

137

138

```
SELECT
CASE
WHEN distancefromhome <= 5 THEN 'Near'
WHEN distancefromhome <= 15 THEN 'Moderate'
ELSE 'Far'
END AS distance_category,
COUNT(*) AS employees_left
FROM employeeattrition
WHERE attrition='Yes'
GROUP BY distance_category;
```

Data Output Messages Notifications

SQL

	distance_category text	employees_left bigint
1	Moderate	70
2	Far	176
3	Near	32

Total rows: 3 Query complete 00:00:00.207

Description

- Employee turnover was examined based on commuting distance from home to workplace.
- The objective was to determine whether travel distance influences employee retention.
- The findings help evaluate the impact of commuting challenges on workforce stability.

Marital Status vs Attrition

 Hr_attrition_analysis/postgres@PostgreSQL 17

    No limit    

Query

Query History

```
137 WHERE attrition='Yes'
138 GROUP BY distance_category;
139
140 SELECT maritalstatus,
141 COUNT(*) AS employees_left
142 FROM employeeattrition
143 WHERE attrition='Yes'
144 GROUP BY maritalstatus;
145
146
```

Data Output

Messages

Notifications

        SQL

	maritalstatus character varying (20) 	employees_left bigint 
1	Married	86
2	Divorced	92
3	Single	100

Total rows: 3 | Query complete 00:00:00.206

Description

- Employee attrition was analyzed across different marital status categories.
- The analysis helps understand whether personal and family-related factors affect retention.
- The results provide additional insights into employee demographics and turnover trends.

Age Group Analysis

Hr_attrition_analysis/postgres@PostgreSQL 17

No limit

Query

Query History

146

147

148

149

150

151

152

153

154

155

156

```
SELECT
CASE
WHEN age BETWEEN 18 AND 25 THEN '18-25'
WHEN age BETWEEN 26 AND 35 THEN '26-35'
WHEN age BETWEEN 36 AND 45 THEN '36-45'
ELSE '46+'
END AS age_group,
COUNT(*) AS employees_left
FROM employeeattrition
WHERE attrition='Yes'
GROUP BY age_group;
```

Data Output

Messages

Notifications

SQL

S

	age_group text	employees_left bigint
1	36-45	73
2	46+	112
3	26-35	66
4	18-25	27

Total rows: 4

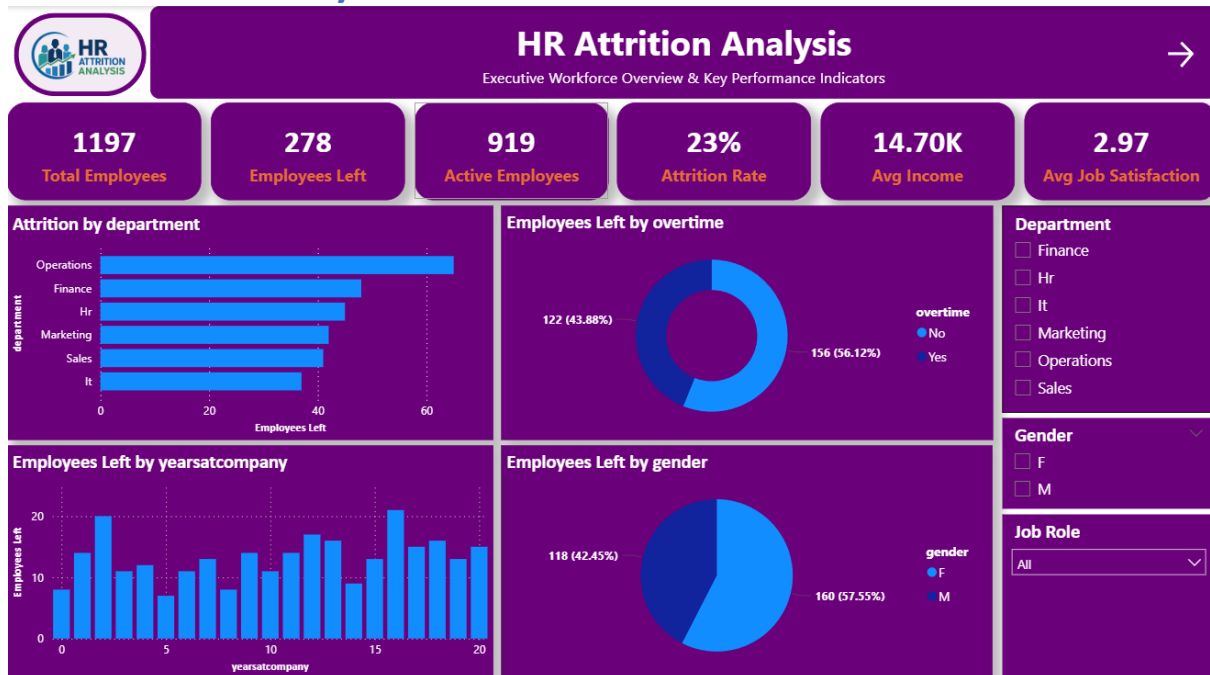
Query complete 00:00:00.347

Description

- Employees were grouped into different age categories for attrition analysis.
- The objective was to identify age groups experiencing higher turnover rates.
- This analysis supports workforce planning and targeted employee engagement initiatives.

Power BI

Executive summary



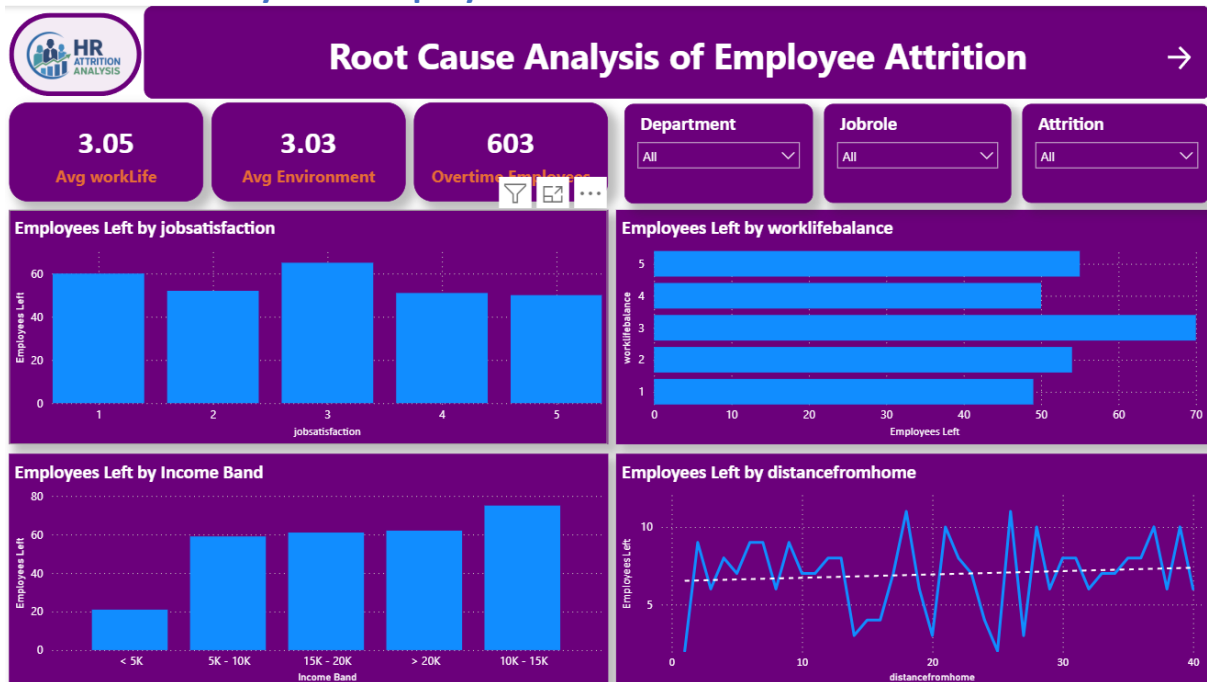
Description:

This dashboard provides an executive overview of the organization's employee attrition. It presents the key HR metrics, including Total Employees, Active Employees, Employees Left, Attrition Rate, Average Monthly Income, and Average Job Satisfaction. The dashboard also visualizes attrition across different departments, overtime status, gender, and years at the company. Interactive slicers allow users to filter the analysis by Department, Gender, and Job Role, helping HR managers quickly identify areas with higher employee turnover.

Business Outcome:

- Provides a high-level view of the organization's workforce.
- Identifies departments with the highest attrition.
- Shows the impact of overtime on employee turnover.
- Helps management monitor overall HR performance.

Root Cause Analysis of Employee Attrition



Description:

This dashboard focuses on identifying the major factors contributing to employee attrition. It displays KPIs such as Average Work-Life Balance, Average Environment Satisfaction, and Total Overtime Employees. Visualizations analyze employee attrition based on Job Satisfaction, Work-Life Balance, Monthly Income Bands, and Distance from Home. Slicers enable users to filter the analysis by Department, Job Role, and Attrition Status to identify patterns across different employee groups.

Business Outcome:

- Identifies the key drivers behind employee attrition.
- Highlights the relationship between overtime, job satisfaction, and work-life balance.
- Supports HR in prioritizing employee retention strategies.
- Enables department-wise analysis of attrition causes.

Workforce Insights & Retention Strategies



Description:

This dashboard provides workforce insights along with actionable recommendations for improving employee retention. It includes KPIs such as Total Departments, Total Job Roles, Average Tenure, and Average Job Satisfaction. The dashboard visualizes employee distribution by Job Role, Age Group, and Environment Satisfaction. A Key Insights section summarizes the major findings from the analysis, highlighting that overtime is the primary contributor to attrition, while lower job satisfaction and poor work-life balance further increase the likelihood of employees leaving the organization.

Business Outcome:

- Presents the overall workforce composition.
- Highlights important workforce trends across age groups and job roles.
- Summarizes the major findings from the analysis.
- Provides recommendations to reduce employee attrition through improved work-life balance and reduced overtime.

Conclusion

The HR Attrition Analysis project was developed to identify the major factors influencing employee attrition and to provide data-driven insights for improving employee retention. The analysis was performed using **Excel** for data cleaning, **PostgreSQL** for data analysis, and **Power BI** for interactive dashboard visualization.

The dashboard revealed that the organization has an **attrition rate of 23%**, with **278 out of 1,197 employees** leaving the company. The analysis identified **overtime** as the most significant factor contributing to employee attrition. Employees with **lower job satisfaction** and **poor work-life balance** were also more likely to leave the organization. Department-wise and demographic analysis further helped identify areas with higher employee turnover.

Based on these findings, the organization can improve employee retention by reducing excessive overtime, promoting a healthier work-life balance, increasing employee engagement, and regularly monitoring employee satisfaction. The interactive dashboard enables HR teams to quickly analyze attrition trends and make informed, data-driven decisions.